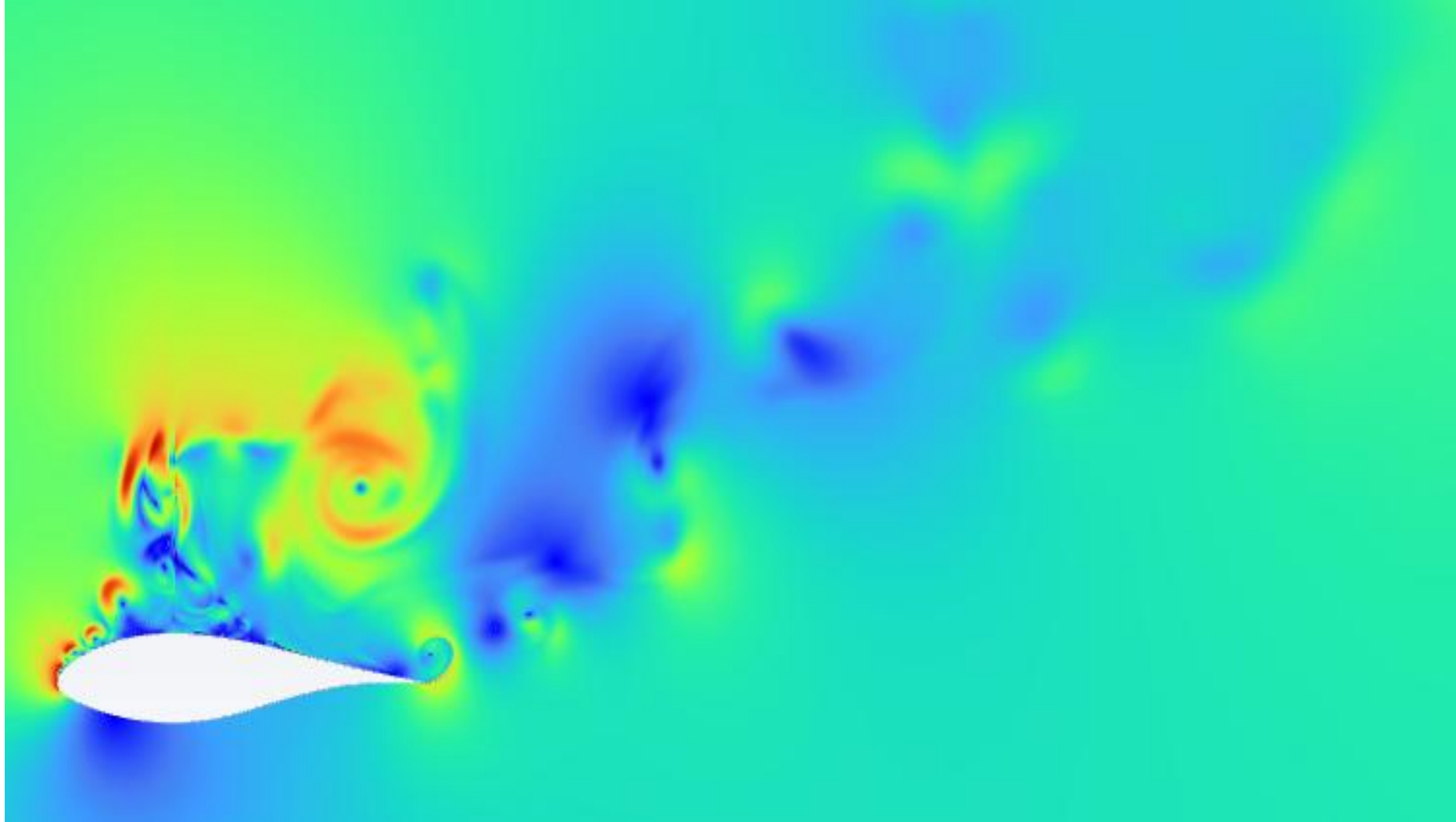


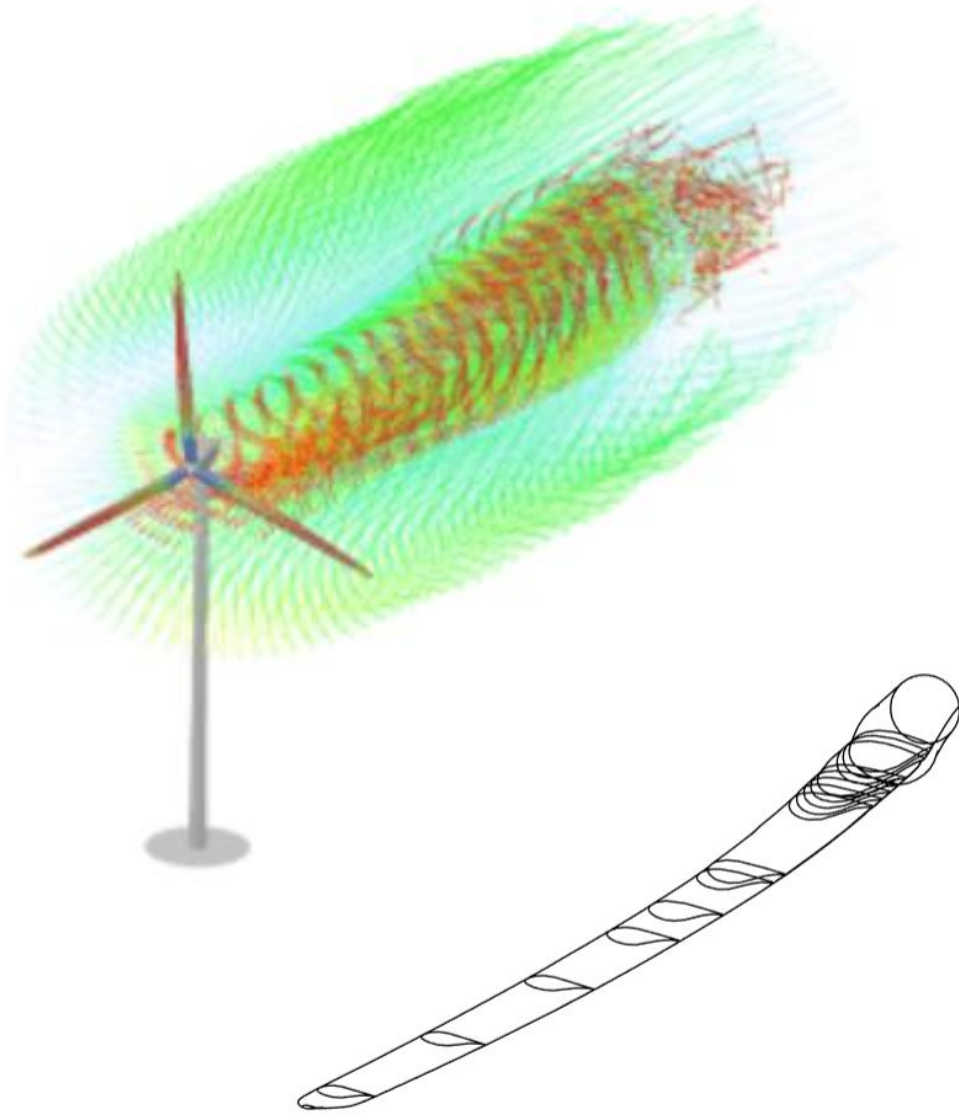
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Abstract

Large offshore wind turbine blades often remain idle in high winds but still experience aerodynamic loading. Alternate vortex shedding from blade airfoil sections generates periodic forces, causing vortex-induced vibrations (VIV) that lead to flap-wise and edgewise oscillations and may reduce structural life. This study examines VIV in an idling blade of a reference 15-MW-240RT offshore wind turbine using coupled aerodynamic-structural analysis and aims to recommend pitch settings that minimize VIV.



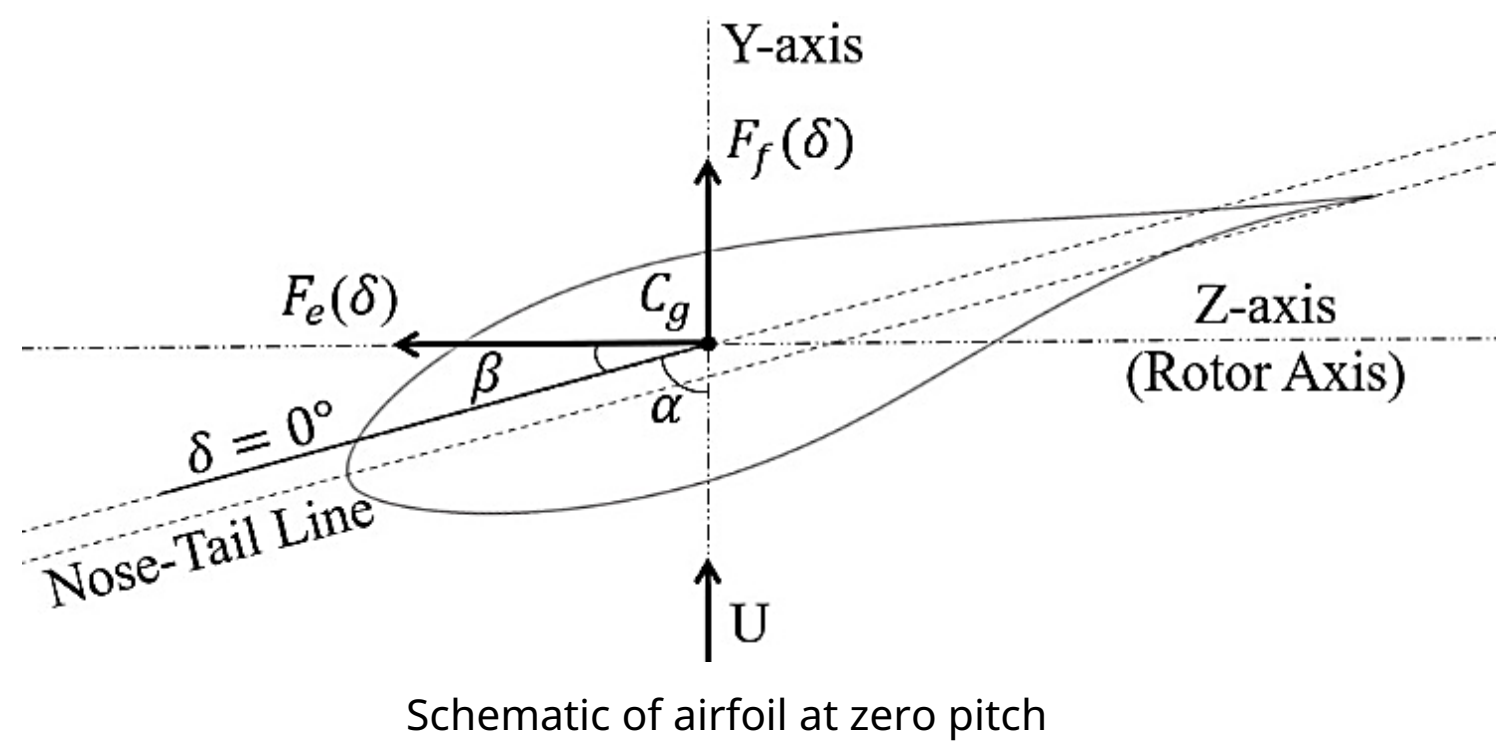
Vortex shedding patterns in velocity contours of FFA-W3-211 airfoil



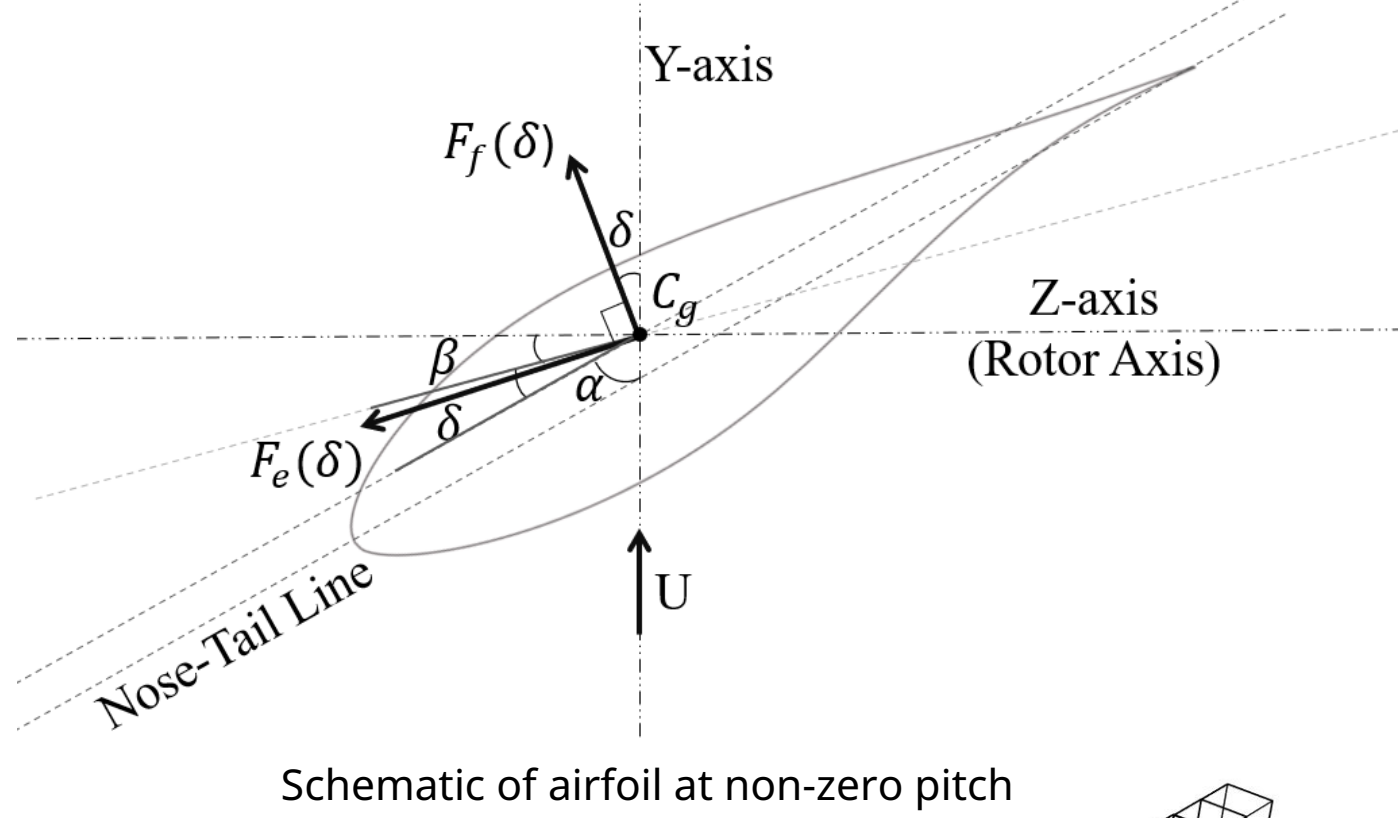
Offshore wind turbine tower pressure contours (above)
Wireframe perspective view of the IEA 15-MW blade (below)

Methodology

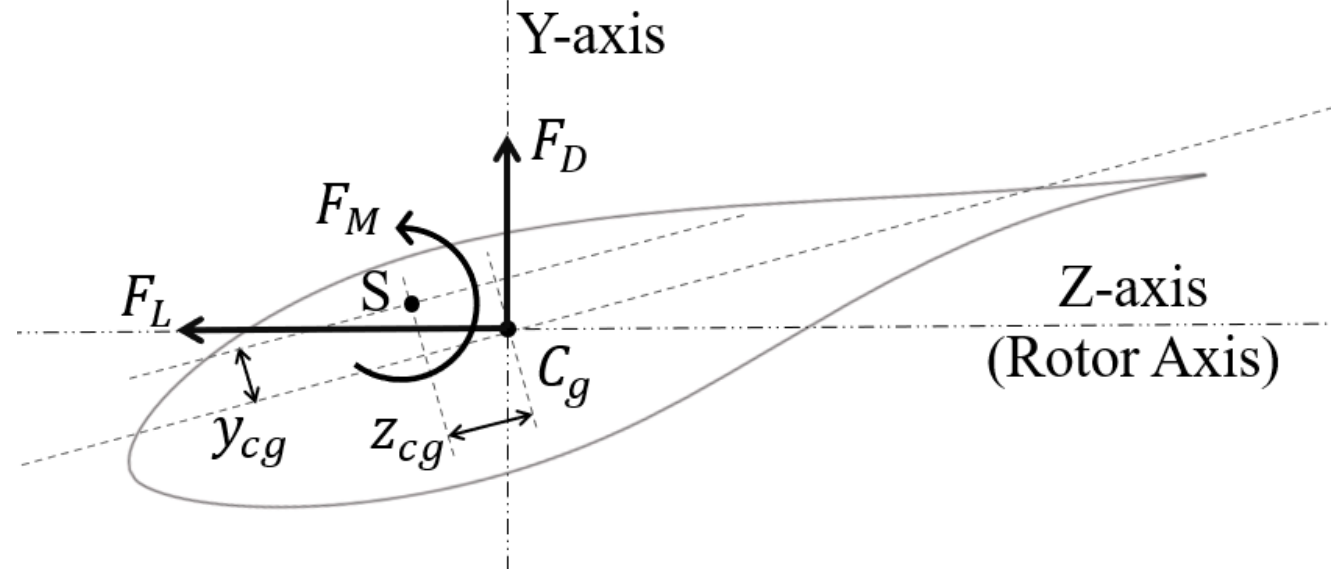
- Blade modelling:** The blade is modeled as a twisted, tapered hollow cantilever beam with spanwise variation in mass and stiffness and coupled flap-wise, edgewise and torsional vibrations.
- Natural frequency analysis:** Free vibration analysis is performed using the Rayleigh-Ritz method to obtain modal frequencies and to check possible resonance with vortex-shedding frequencies.
- Aerodynamic excitation:** 2D CFD simulations using URANS k-w SST model (ANSYS Fluent) are performed for seven FFA-W3 airfoil sections using dynamic meshing to obtain lift coefficient (Cl), drag coefficient (Cd), moment coefficient (Cm) and vortex-shedding frequency.
- Structural Analysis:** Aerodynamic forces are resolved into flap-wise and edge-wise excitation components considering blade pitch. Modal superposition and numerical integration are used to compute dynamic blade response and stresses.
- Parametric study:** Wind speed is varied as 3,7,10,59,15 and 25m/s while pitch angle is varied from 105°-60° to quantify relationship between VIV behavior and root stresses at these configurations.
- AI/ML prediction:** Machine learning techniques are used to predict root stress values for intermediate wind and pitch conditions, enabling smart and efficient evaluation of safe operating pitch angles.



Schematic of airfoil at zero pitch



Schematic of airfoil at non-zero pitch

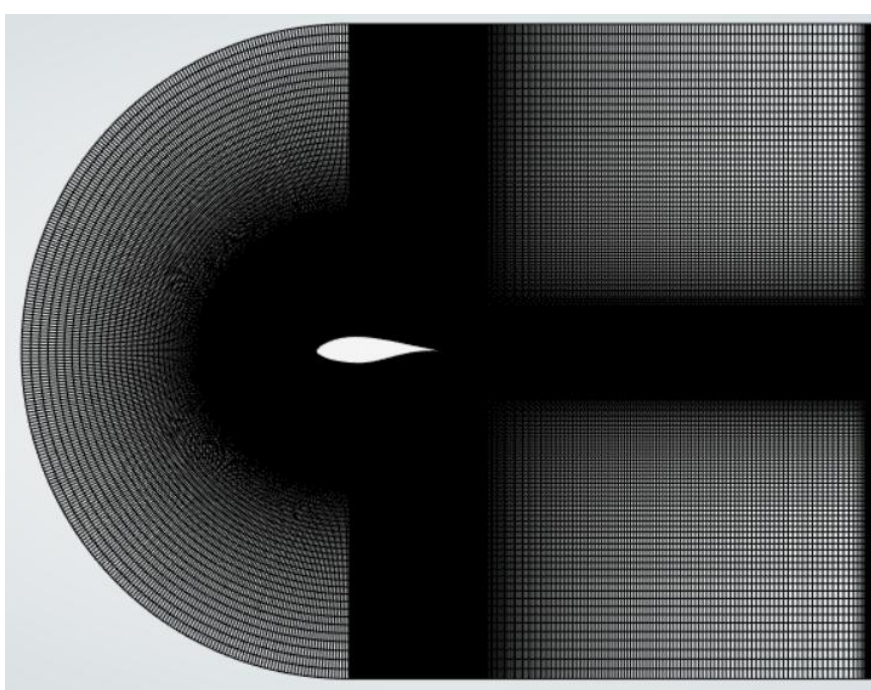


Schematic of airfoil with force vectors

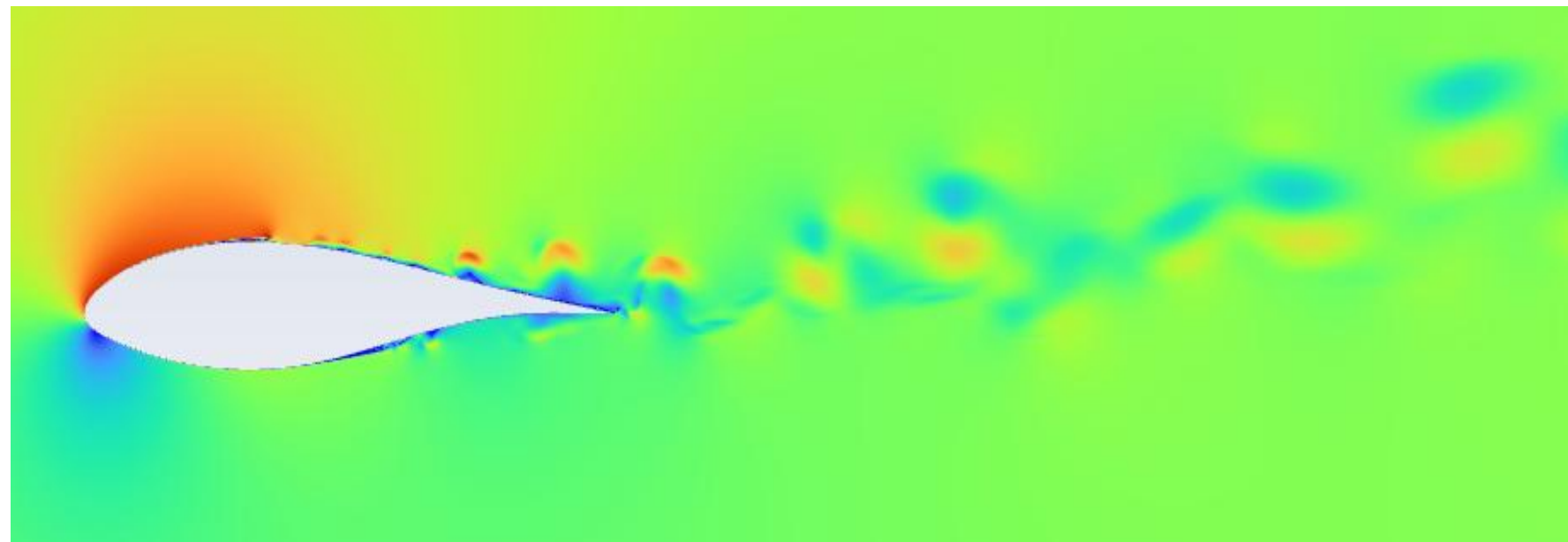
#	Blade Section	$c(\bar{x})$	$x \downarrow$	$\delta = 105^\circ$	$\delta = 95^\circ$	$\delta = 85^\circ$	$\delta = 75^\circ$	$\delta = 60^\circ$
1	SNL-FFA-W3-500	5.73	19.10	-23.67	-13.67	-3.67	6.33	21.33
2	FFA-W3-360	5.39	31.04	-20.66	-10.66	-0.66	9.34	24.34
3	FFA-W3-330blend	4.77	40.59	-18.48	-8.48	1.52	11.52	26.52
4	FFA-W3-301	4.21	52.53	-16.84	-6.84	3.16	13.16	28.16
5	FFA-W3-270blend	3.71	64.46	-15.64	-5.64	4.36	14.36	29.36
6	FFA-W3-241	3.20	76.40	-14.41	-4.41	5.59	15.59	30.59
7	FFA-W3-211	2.22	90.73	-13.22	-3.22	6.78	16.78	31.78

Angle of attack as a function of pitch and spanwise location

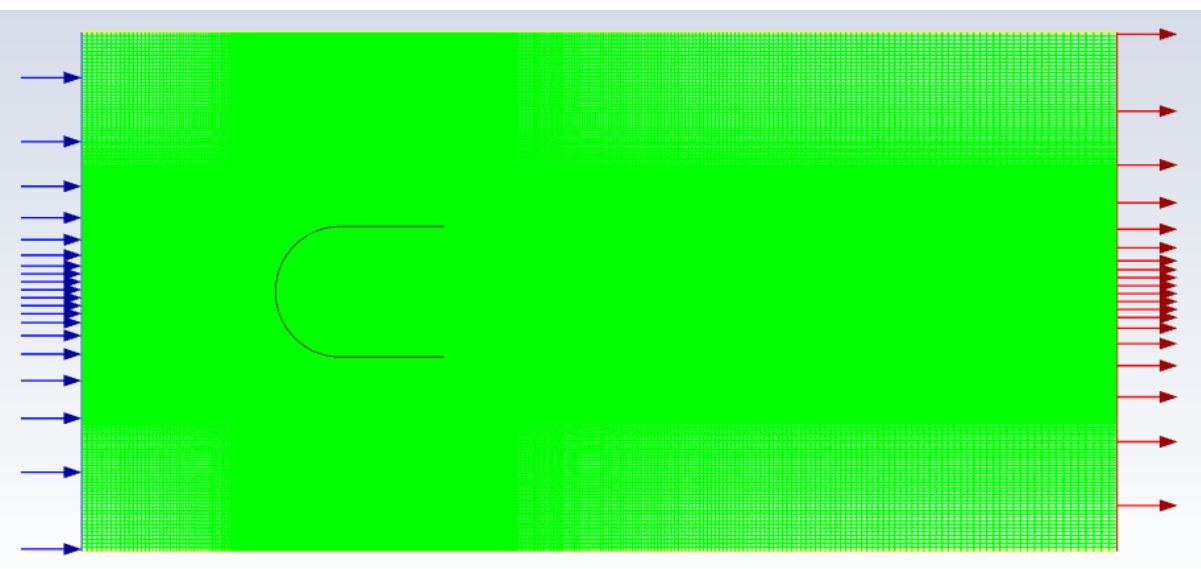
Wireframe view of modelled blade and Top view of airfoil profiles



View of structured airfoil mesh



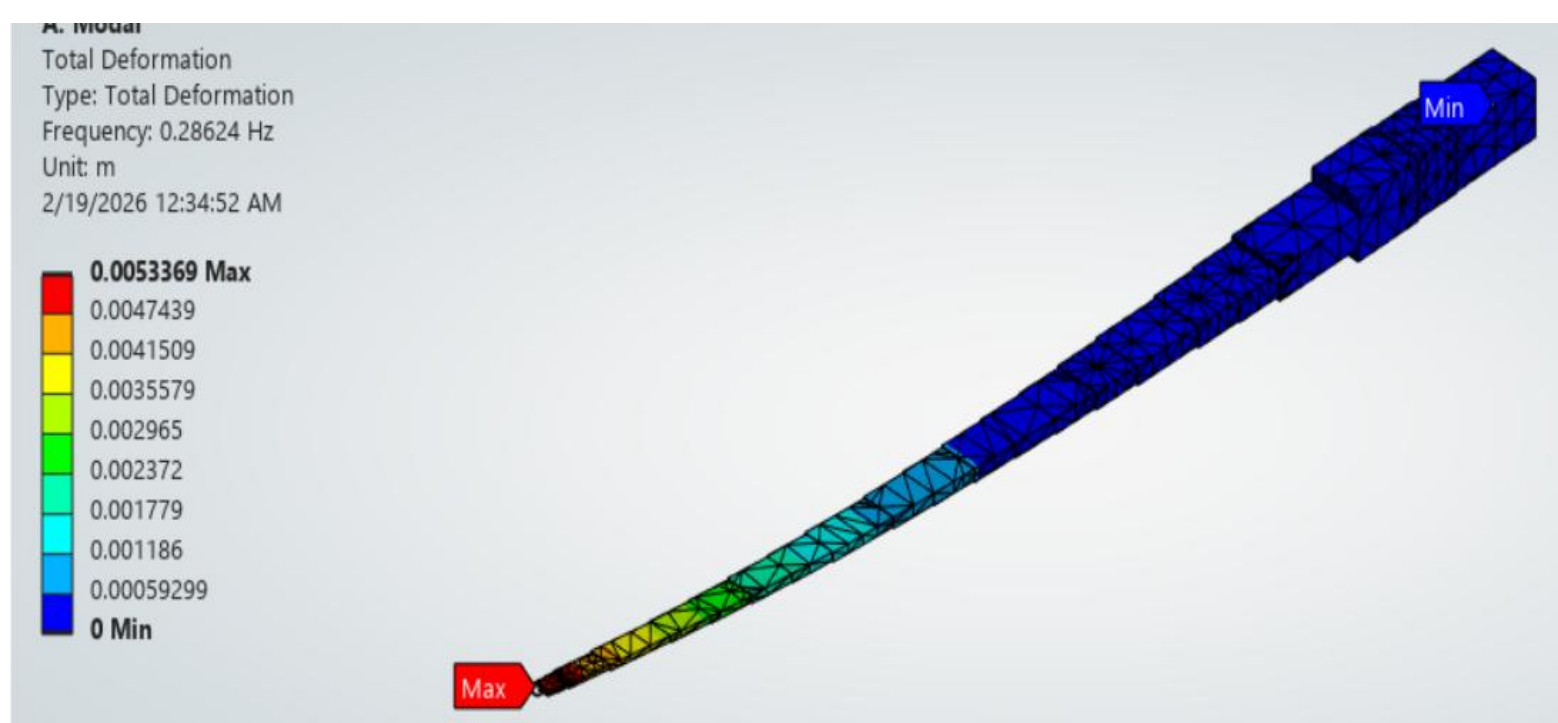
VIV patterns of FFA-W3-241 airfoil at 8m/s wind speed and 10° AOA



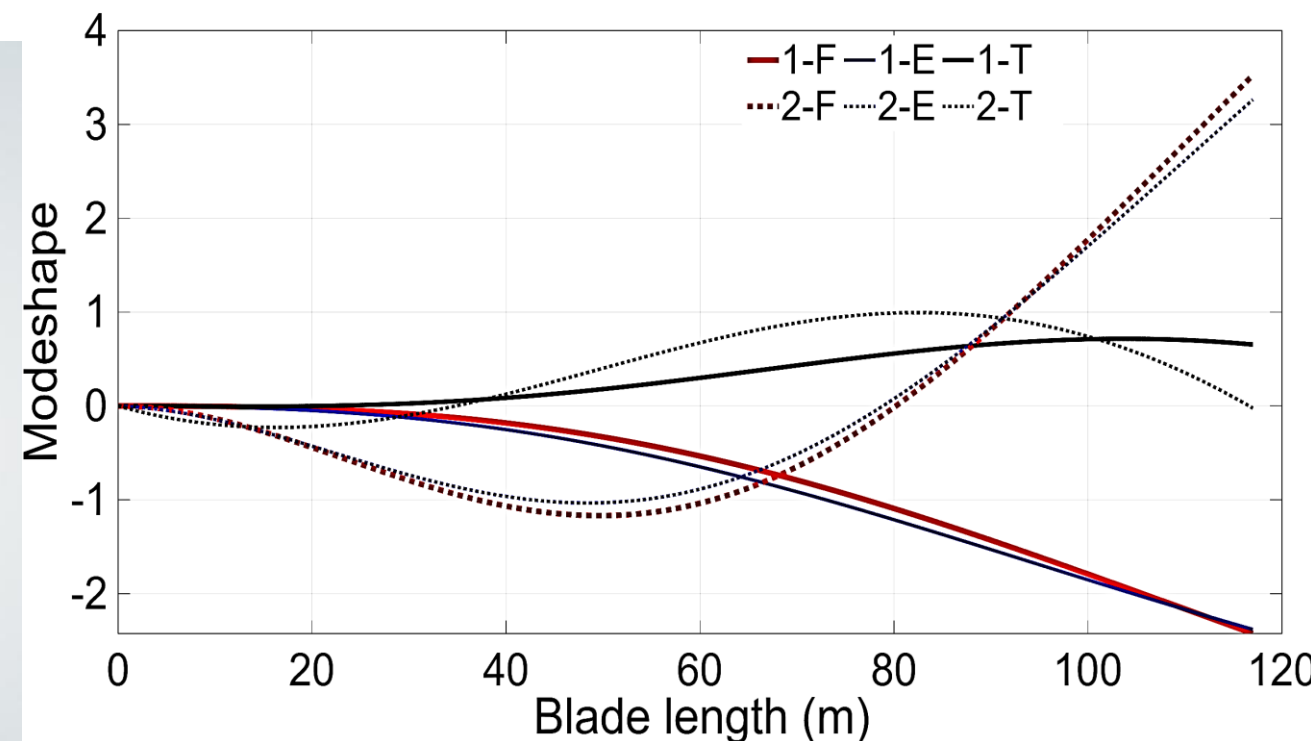
View of dynamic meshing setup in Fluent

Results

- Free vibration analysis** shows the first coupled flap-wise and edgewise frequencies to be around ~0.52-0.78 Hz, thus verifying the structural setup.
- Tuning factor analysis** filters the spanwise locations and their wind, pitch configurations where the possibility of VIV amplification exists.
- CFD simulations** confirm and quantify VIV parameters (Cl,Cd,Cm and f_viv) for different airfoil sections and configurations.
- Structural results** reveal root bending stresses as a function of pitch angles for given wind speeds.
- ML models** trained on the result datasets obtained predict root stresses for intermediate wind speed and pitch values.
- Final analysis** reveals the optimum pitch angles for minimum VIV caused root stresses at different wind speeds.



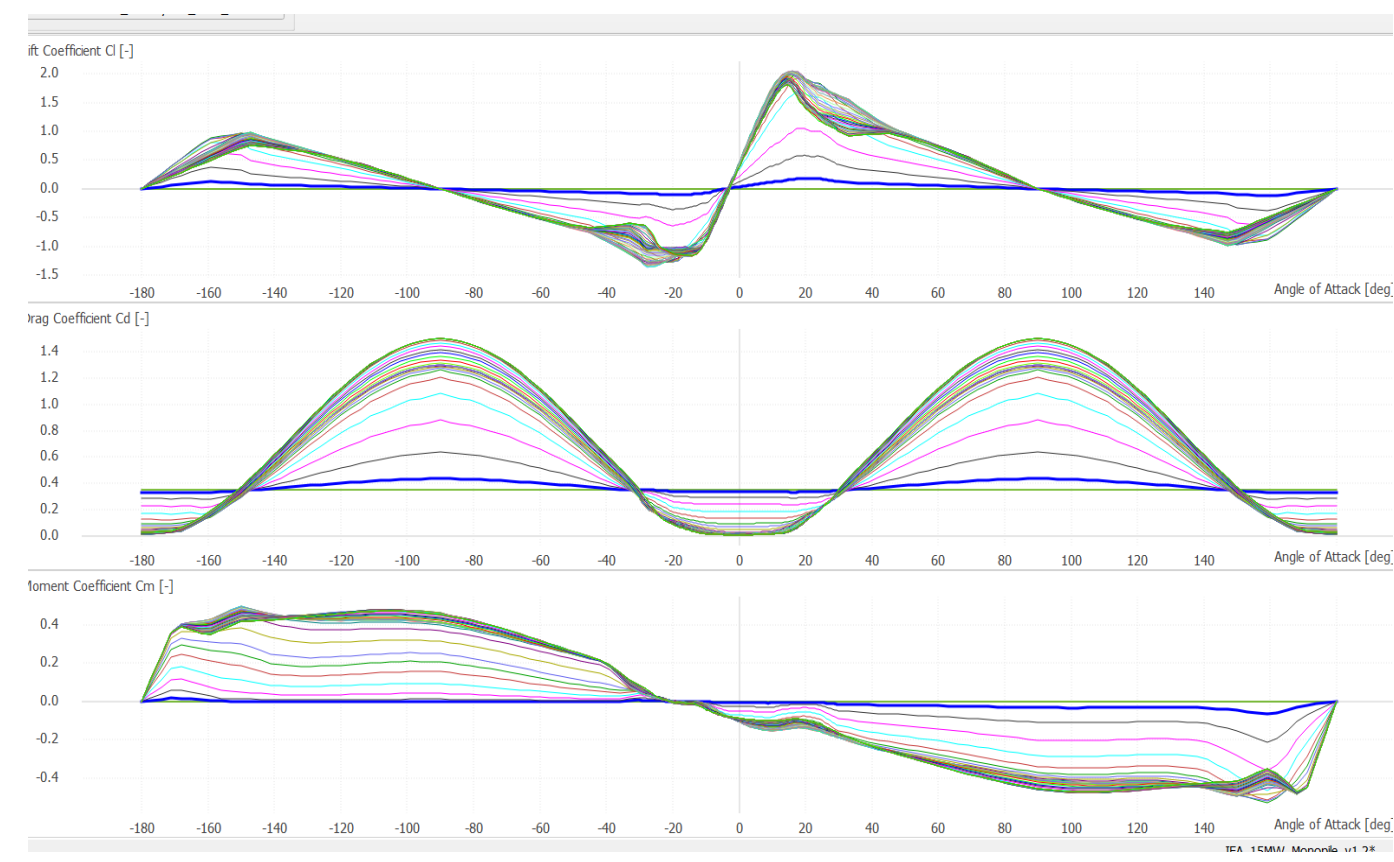
Blade modelled in ANSYS Mechanical



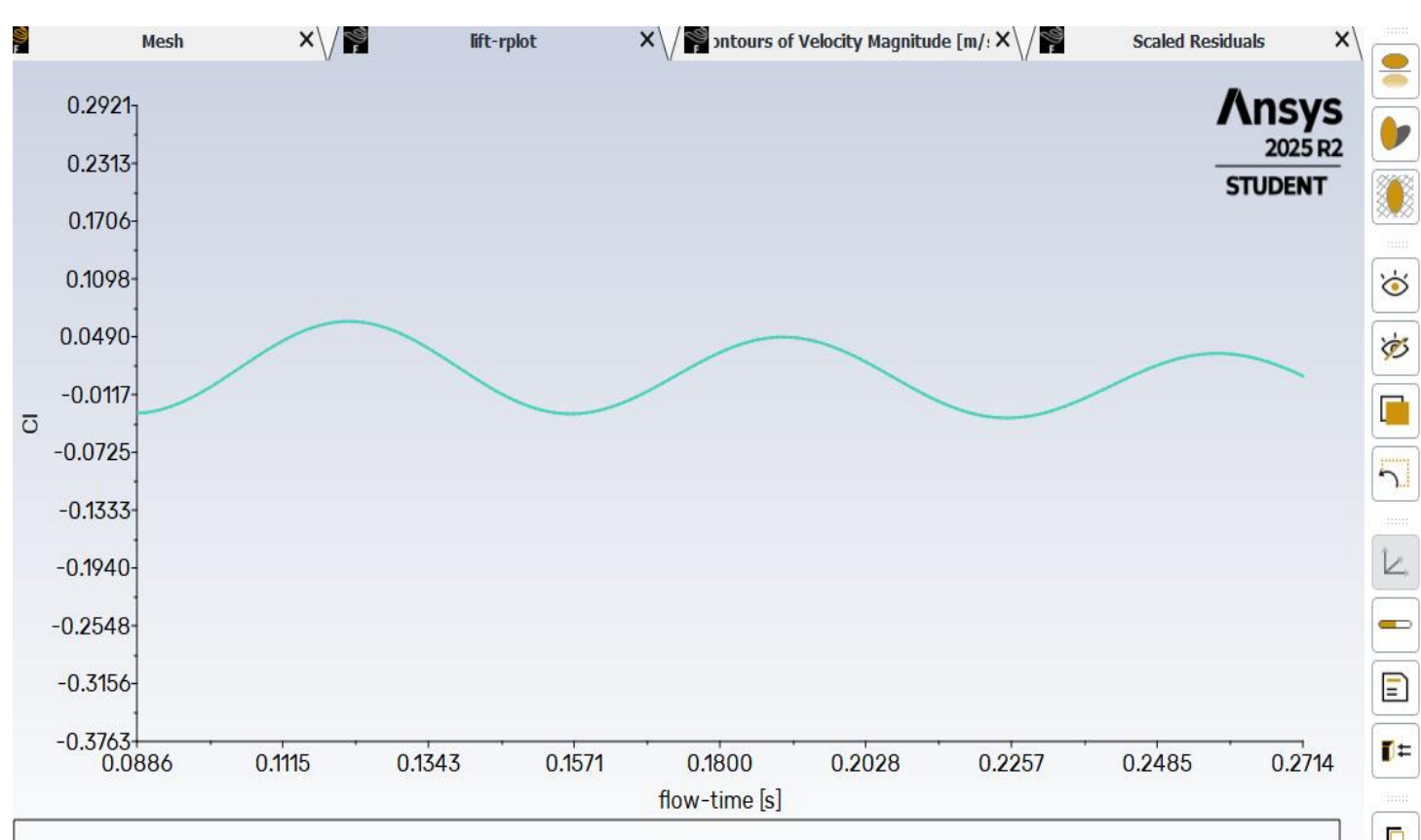
1st and 2nd Natural modeshapes

	$\delta = 105^\circ$			$\delta = 95^\circ$			$\delta = 85^\circ$			$\delta = 75^\circ$			$\delta = 60^\circ$		
Aerofoil	1F	1E	1T	1F	1E	1T	1F	1E	1T	1F	1E	1T	1F	1E	1T
Circular	0.76	0.53	0.31	0.76	0.53	0.31	0.76	0.53	0.31	0.76	0.53	0.31	0.76	0.53	0.31
SNL-FFA-W3-500	0.71	0.50	0.29	1.07	0.75	0.43	1.42	1.00	0.58	1.42	1.00	0.58	0.71	0.50	0.29
FFA-W3-360	0.76	0.53	0.31	1.13	0.80	0.46	1.51	1.06	0.61	1.51	1.06	0.61	0.76	0.53	0.31
FFA-W3-330blend	1.28	0.90	0.52	1.28	0.90	0.52	1.71	1.20	0.69	1.71	1.20	0.69	0.85	0.60	0.35
FFA-W3-301	1.45	1.02	0.59	1.45	1.02	0.59	1.93	1.36	0.79	1.45	1.02	0.59	0.97	0.68	0.39
FFA-W3-270blend	1.65	1.16	0.67	1.65	1.16	0.67	2.20	1.54	0.89	1.65	1.16	0.67	1.10	0.77	0.45
FFA-W3-241	1.91	1.34	0.77	2.54	1.79	1.03	2.54	1.79	1.03	1.91	1.34	0.77	1.27	0.89	0.52
FFA-W3-211	2.76	1.94	1.12	3.68	2.58	1.49	3.68	2.58	1.49	2.76	1.94	1.12	1.84	1.29	0.75

Tuning Factor table for rated wind speed 10.59m/s



Static lift, drag and moment coefficients obtained from QBlade

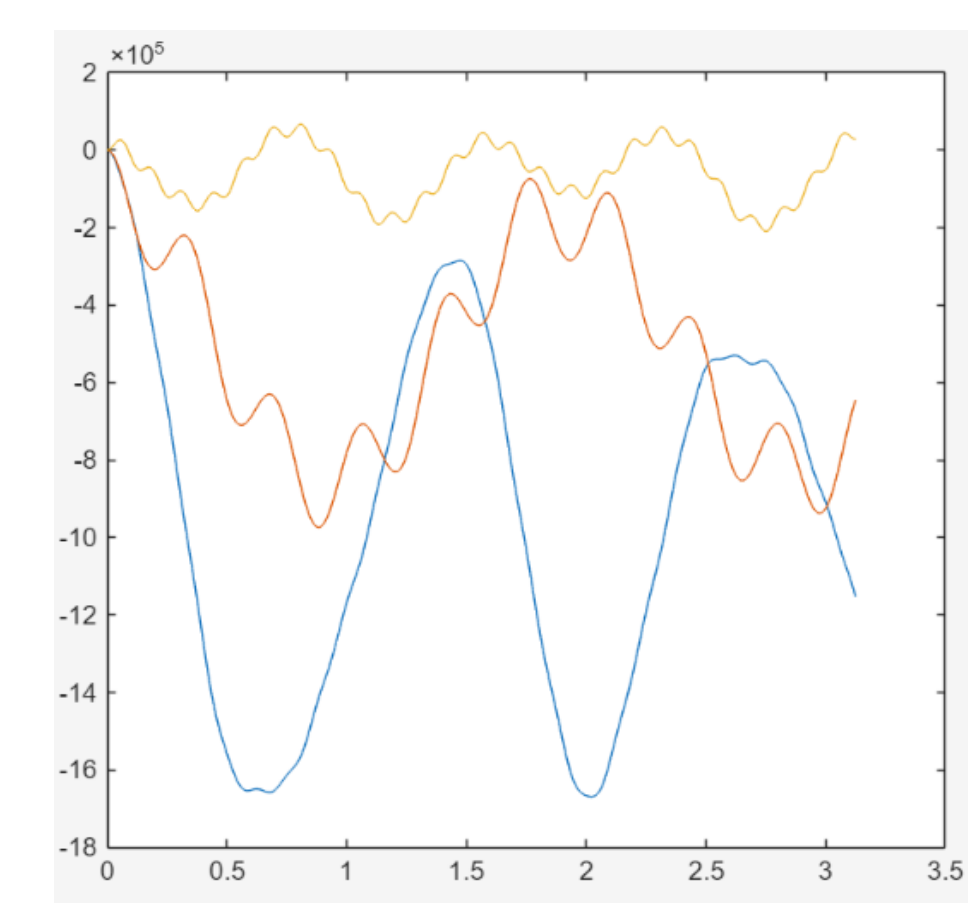


Dynamic lift coefficient obtained from ANSYS Fluent

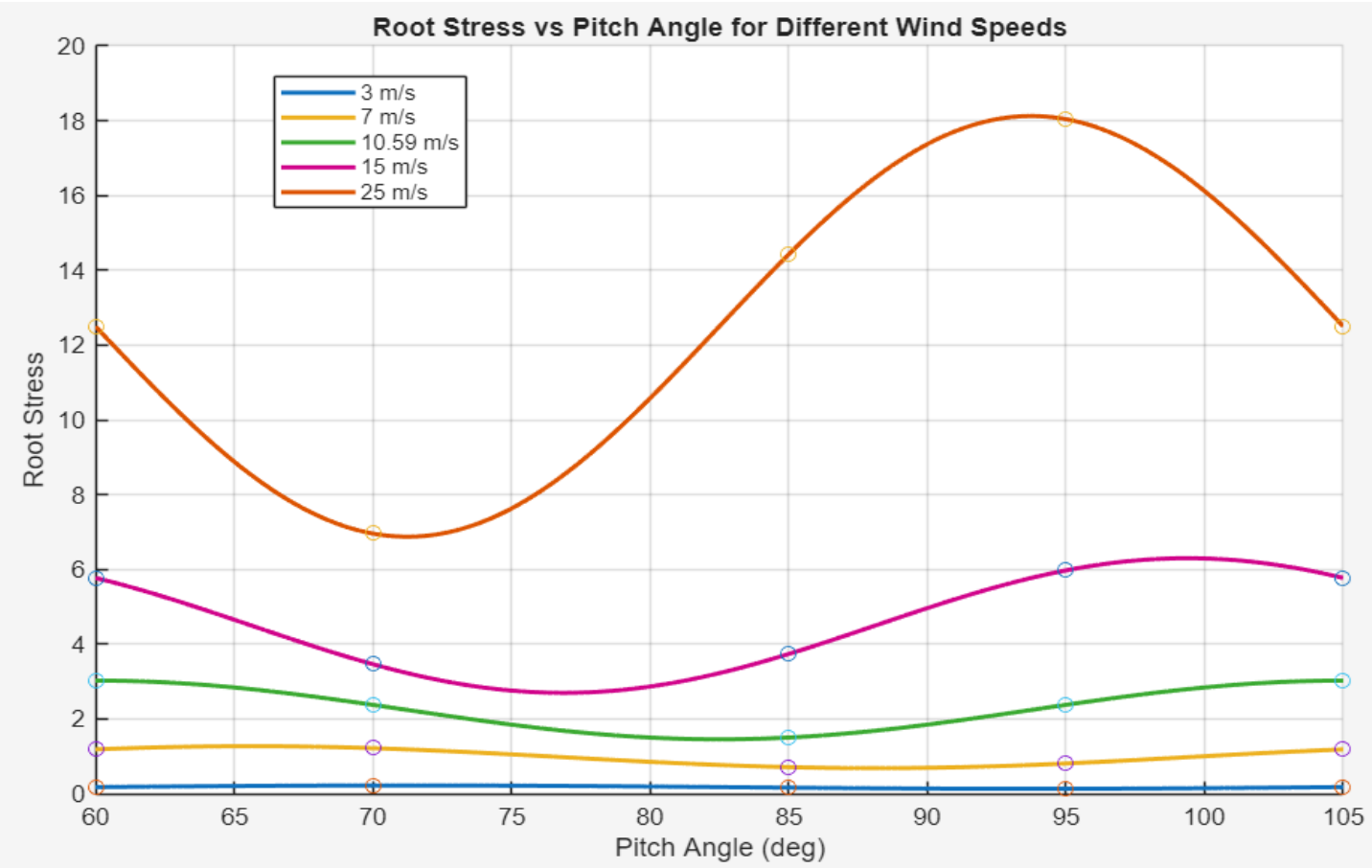
Conclusion

The results demonstrate that VIV risk depends strongly on the chosen pitch angle during shutdown and simple pitch control strategies can reduce fatigue and increase the blade life significantly as compared to conventional feathering strategies. Furthermore, the study establishes a clear coupling between aerodynamic excitation and structural response, highlighting the importance of integrated aeroelastic analysis in predicting VIV behavior. The incorporation of machine learning enhances the efficiency of the framework by enabling rapid prediction of intermediate operating conditions without the need for exhaustive simulations.

Root bending moments as function of time

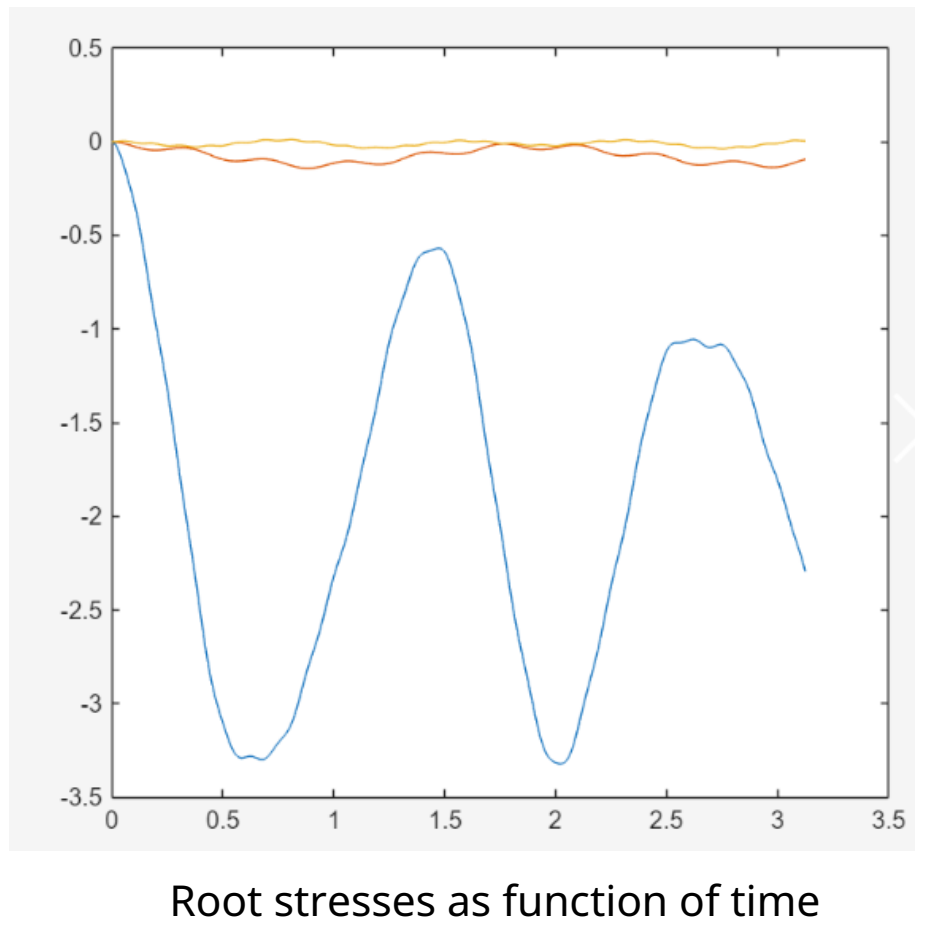


Root stress from VIV as a function of pitch at different wind speeds



Station	Aerofoil	x	Im/s	ckl	Twist(deg)	Pitch	AOA(deg)	Re (10 ⁵)	St	Γ_{tip}	Cl	Cd	Cm	L	D	T
1	Circular	0.000	5.200	18.59	105	-30.59	3.44	0.18	0.035	0.0001	0.35	-0.0001	0.000319	1.11475	-0.00186	
2	Circular	2.388	5.209	15.59	105	-30.59	3.45	0.18	0.035	0.0001	0.35	-0.0001	0.000319	1.11645	-0.00186	
3	Circular	4.776	5.238	15.41	105	-30.41	3.47	0.18	0.034	0.0001	0.35	-0.0001	0.000321	1.122872	-0.00186	
4	Circular	7.163	5.293	14.95	105	-29.95	3.51	0.18	0.034	0.0001	0.35	-0.0001	0.000324	1.134757	-0.00172	
5	Circular	9.551	5.367	14.26	105	-29.26	3.55	0.18	0.034	0.0001	0.35	-0.0001	0.000328	1.150623	-0.00176	
6	Circular	11.939	5.452	13.40	105	-28.40	3.61	0.18	0.033	0.0001	0.35	-0.0001	0.000334	1.168792	-0.00182	
7	Circular	14.327	5.540	12.42	105	-27.42	3.67	0.18	0.032	0.0001	0.35	-0.0001	0.000339	1.187444	-0.00188	
8	Circular	16.714	5.622	11.39	105	-26.39	3.72	0.18	0.032	0.0001	0.35	-0.0001	0.000344	1.205179	-0.00194	
9	FFA-W3-500	19.102	5.693	10.37	105	-25.37	3.77	0.18	0.032	-1.27443	0.181004	-0.00413	-4.44352	0.561367	-0.08198	
10	FFA-W3-500	21.490	5.743	9.40	105	-24.40	3.80	0.18	0.031	-1.28015	0.151291	-0.00962	-4.43227	0.532144	-0.19405	
11	FFA-W3-500	23.878	5.785	8.55	105	-23.55	3.82	0.18	0.031	-1.28773	0.145127	-0.01148	-4.4057	0.514207	-0.2038	
12	FFA-W3-500	26.265	5.756	7.83	105	-22.83	3.81	0.18	0.031	-1.22688	0.140985	-0.01289	-4.32553	0.487059	-0.26168	
13	FFA-W3-500	28.653	5.703	7.19	105	-22.19	3.78	0.18	0.032	-1.19761	0.136516	-0.01399	-4.18341	0.476872	-0.2787	
14	FFA-W3-360	31.041	5.629	6.55	105	-21.55	3.71	0.18	0.032	-1.05959	0.109053	-0.01007	-3.63881	0.376562	-0.19379	
15	FFA-W3-360	33.429	5.472	5.93	105	-20.93	3.62	0.18	0.033	-1.0347	0.075525	-0.01107	-3.46763	0.25311	-0.20308	
16	FFA-W3-360	35.816	5.323	5.35	105	-20.35	3.53	0.18	0.034	-1.01243	0.070963	-0.01155	-3.30079	0.231354	-0.20043	
17	FFA-W3-360	38.204	5.165	4.80	105	-19.80	3.42	0.18	0.035	-0.9913	0.066722	-0.01194	-3.13851	0.211139	-0.19526	
18	FFA-W3-330	40.592	5.019	4.30	105	-19.30	3.32	0.18	0.036	-1.04172	0.057232	-0.00403	-3.20265	0.179553	-0.06219	
19	FFA-W3-330	42.980	4.886	3.85	105	-18.85	3.24	0.18	0.037	-1.03353	0.053871	-0.00393	-3.09291	0.161213	-0.05743	
20	FFA-W3-330	45.367	4.768	3.45	105	-18.45	3.16	0.18	0.038	-1.02682	0.050851	-0.00372	-2.9907	0.148795	-0.05198	
21	FFA-W3-330	47.755	4.655	3.08	105	-18.08	3.08	0.18	0.039	-1.02211	0.04833	-0.00346	-2.91109	0.137798	-0.04696	
22	FFA-W3-330	50.143	4.541	2.73	105	-17.73	3.01	0.18	0.040	-1.01578	0.045889	-0.00322	-2.82526	0.127834	-0.04066	
23	FFA-W3-301	52.531	4.428	2.41	105	-17.41	2.93	0.18	0.041	-1.00951	0.043541	-0.00305	-2.748105	0.096397	-0.04863	
24	FFA-W3-301	54.918	4.317	2.11	105	-17.11	2.86	0.18	0.042	-1.00364	0.041373	-0.00286	-2.67856	0.085771	-0.06117	
25	FFA-W3-301	57.306	4.208	1.83	105	-16.83	2.79	0.18	0.043	-1.07958	0.033717	-0.00269	-2.61758	0.081745	-0.07151	
26	FFA-W3-301	59.694	4.102	1.56	105	-16.56	2.72	0.18	0.044	-1.05635	0.030422	-0.00284	-2.65384	0.076429	-0.09627	

Example database used to train ML model



Root stresses as function of time